



JOHNS HOPKINS
BLOOMBERG SCHOOL
of PUBLIC HEALTH

Measures of Disease Burden

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Lecture Objectives

- ▶ Describe disease burden using the following measures:
 - ▶ Mortality rate (MR)
 - ▶ Years of potential life lost (YPLL)
 - ▶ Life expectancy (LE)
 - ▶ Disability-adjusted life year (DALY)
 - ▶ Quality-adjusted life year (QALY)
- ▶ To calculate and interpret:
 - ▶ Direct age-adjusted mortality rates
 - ▶ Indirect age-adjusted mortality rates

Recommended Readings

- ▶ Tutorials from the National Association for Public Health Statistics and Information Systems (NAPHSIS)
 - ▶ Direct age adjustment
 - ▶ Indirect age adjustment
 - ▶ YPLL
- ▶ Stevens GA, et al., 2016, Guidelines for accurate and transparent health estimates reporting: the GATHER statement, *Lancet*,
[https://doi.org/10.1016/S0140-6736\(16\)30388-9](https://doi.org/10.1016/S0140-6736(16)30388-9)

Mortality Rates (MRs)—1

“Life is a sexually transmitted disease and the mortality rate is 100%.”

—R.D. Laing
(1927–1989)

Mortality Rates (MRs)—2

- ▶ MR is the rate of death in a population

$$\text{MR} = \frac{\text{Number of deaths}}{\text{Number of people at risk for death}} * 1,000$$

- ▶ MRs are typically expressed as deaths per 1,000 or 100,000 ***people*** or ***population per year***

Mortality Rate: Characteristics—1

- ▶ MRs are often specified by descriptions of the subpopulation for which they are calculated
 - ▶ Person
 - ▶ Place
 - ▶ Time
- ▶ MRs can be used to monitor trends
 - ▶ Cause-specific MRs are a marker of the severity of illness
- ▶ MRs can be estimated at the local, state, and national level

Mortality Rate: Characteristics—2

- ▶ MRs are dependent upon age and gender distributions of the population
 - ▶ Age is one of the strongest predictors of death in a population
- ▶ MRs are adjusted for age through ***direct or indirect age adjustment (or standardization)*** when MRs in one group are compared to MRs in another group
 - ▶ To control for the effects of differences in population age distributions

Direct age adjustment

Skills-building Example

Direct Age Standardization

- Direct age standardization applies the MRs from the population of interest to a **standard population** to calculate the **expected number of deaths in the standard population**,
 - ▶ The expected number of deaths are then summed to get the age-adjusted MRs
 - ▶ The age-adjusted MRs are comparable to other populations because the effect of the age distribution of the population of interest is removed in the age-adjusted MR

The Standard Population

- For US comparisons
 - ▶ **2000 US standard population (1998)**
 - Replaced the 1940 US standard population, the 1970 civilian non-institutional population and the 1980 US resident population
 - More older individuals as the population “aged”
- For International comparisons
 - ▶ **WHO standard population** is based on the world average (2001) population between 2000-2025
 - Terminal age group is ≥ 100 years
 - Has fewer children and more adults aged ≥ 70 years than the **Segi world standard**
 - Younger than the **Scandinavian (European) standard**

Anderson, et al. Age Standardization of Death Rates: Implementation of the Year 2000 Standard. National Vital Statistics Reports 1998:47(3):1-17.

Ahmad, et al. Age Standardization of Rates: A New WHO Standard. GPE Discussion Paper Series: No 31., 2001.

Mortality Racial/Ethnic Disparities, Maryland, 2013

- Example: Investigate disparities in mortality rates by race in Maryland
- Data
 - ▶ Maryland Vital Statistics Annual Report 2013 (<http://dhmh.maryland.gov/vsa/documents/13annual.pdf>)
 - Numbers of deaths (*Table 41*)
 - Population at risk for death (*Table 3*)
 - ▶ Data are stratified by age, sex, and race/ethnicity

Number of Deaths by Race/Ethnicity, Maryland, 2013

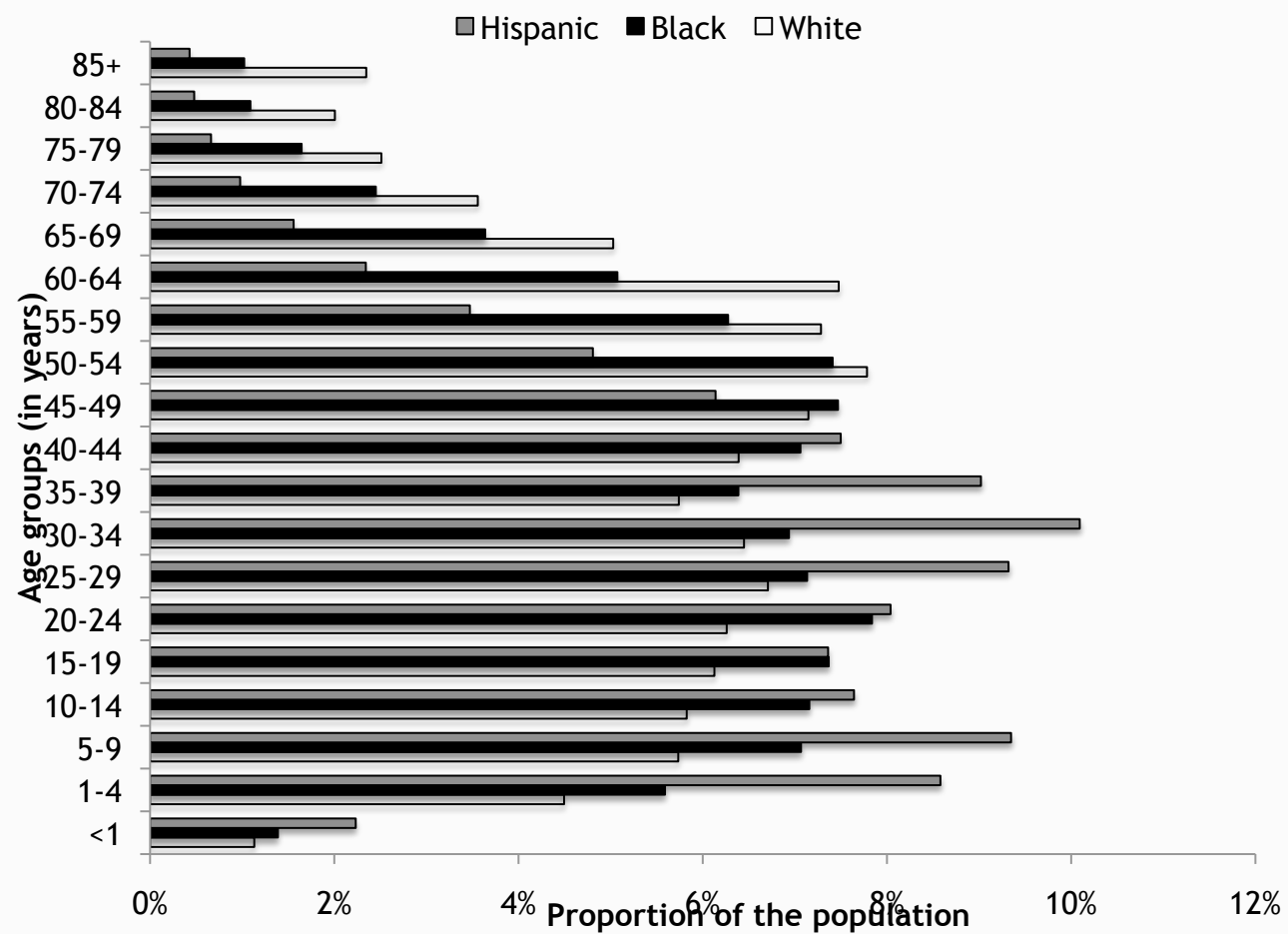
Age (in years)	White	Black	Hispanic
<1	150	250	49
1-4	26	34	8
5-14	29	36	7
15-24	263	229	31
25-34	461	367	50
35-44	609	499	65
45-54	1,801	1,344	69
45-64	3,561	2,300	100
65-74	5,071	2,488	102
75-84	7,718	2,444	111
≥85	11,643	2,364	149

Data Source: Maryland Vital Statistics Annual Report, 2014. Table 41. (<http://dhmh.maryland.gov/vsa/documents/13annual.pdf>)

Age Distributions by Race/Ethnicity, Maryland, 2013 - I

Age (in years)	White		Black		Hispanic	
	n	%	n	%	n	%
<1	41,877	1%	25,599	1%	11,887	2%
1-4	166,391	4%	103,105	6%	45,679	9%
5-9	212,268	6%	130,343	7%	49,762	9%
10-14	215,669	6%	132,004	7%	40,685	8%
15-19	226,796	6%	135,886	7%	39,188	7%
20-24	231,746	6%	144,555	8%	42,798	8%
25-29	248,308	7%	131,567	7%	49,616	9%
30-34	238,696	6%	127,928	7%	53,731	10%
35-39	212,503	6%	117,798	6%	48,021	9%
40-44	236,535	6%	130,226	7%	39,918	7%
45-49	264,579	7%	137,743	7%	32,683	6%
50-54	288,128	8%	136,673	7%	25,593	5%
55-59	269,632	7%	115,737	6%	18,483	3%
60-64	276,753	7%	93,572	5%	12,470	2%
65-69	186,129	5%	67,108	4%	8,297	2%
70-74	131,710	4%	45,203	2%	5,207	1%
75-79	92,967	3%	30,347	2%	3,523	1%
80-84	74,275	2%	20,093	1%	2,550	0%
85+	86,883	2%	18,850	1%	2,283	0%

Age Distributions by Race/Ethnicity, Maryland, 2013 - II



Crude MR per 1,000

White
Both Sexes

WORK IT OUT >>

Expected
number
of deaths
(cross
product)

Age (in years)	Deaths	Population	Crude MR per 1000	US 2000 Std Pop	
<1	150	41,877		0.013818	
1-4	26	166,391		0.055317	
5-9		212,268			
10-14		215,669			
5-14	29	427,937		0.145565	
15-19		226,796			
20-24		231,746			
15-24	263	458,542		0.138646	
25-29		248,308			
30-34		238,696			
25-34	461	487,004		0.135573	
35-39		212,503			
40-44		236,535			
35-44	609	344,370		0.162613	
45-49		264,579			
50-54		288,128			
45-54	1,801	406,592		0.134834	

Crude MR per 1,000

*MR=# of deaths / population at risk
='Deaths' / 'Population' * 1000
=B10/C10*1000 (spreadsheet
program
for age <1 year)*

**White
Both Sexes**

**Expected
number
of deaths
(cross
product)**

Age (in years)	Deaths	Population	Crude MR per 1000	US 2000 Std Pop	Expected number of deaths (cross product)
<1	150	41,877	3.58	0.013818	
1-4	26	166,391	0.16	0.055317	
5-9		212,268			
10-14		215,669			
5-14	29	427,937	0.07	0.145565	
15-19		226,796			
20-24		231,746			
15-24	263	458,542	0.57	0.138646	
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40-44		236,535			
35-44	609	344,370	1.77	0.162613	
45-49		264,579			
50-54		288,128			
45-54	1,801	406,592	4.43	0.134834	

Expected Number of Deaths (Cross Product)

White
Both Sexes

WORK IT OUT >>

Age (in years)	Deaths	Population	Crude MR per 1000	US 2000 Std Pop	Expected number of deaths (cross product)
<1	150	41,877	3.58	0.013818	
1-4	26	166,391	0.16	0.055317	
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35-44	609	344,370	1.77	0.162613	
45-49		264,579			
50-54		288,128			
45-54	1,801	406,592	4.43	0.134834	

Expected Number of Deaths (Cross Product)

= 'Crude MR per 1000' * 'US 2000 Std Pop'
= D10/E10 (spreadsheet program
for age <1 year)

White Both Sexes					
Age (in years)	Deaths	Population	Crude MR per 1000	US 2000 Std Pop	Expected number of deaths (cross product)
<1	150	41,877	3.58	0.013818	0.05
1-4	26	166,391	0.16	0.055317	0.01
5-9		212,268			
10-14		215,669			
5-14	29	427,937	0.07	0.145565	0.01
15-19		226,796			
20-24		231,746			
15-24	263	458,542	0.57	0.138646	0.08
25-29		248,308			
30-34		238,696			
25-34	461	487,004	0.95	0.135573	0.13
35-39		212,503			
40-44		236,535			
35-44	609	344,370	1.77	0.162613	0.29
45-49		264,579			
50-54		288,128			
45-54	1,801	406,592	4.43	0.134834	0.60

Age-adjusted Mortality Rate

White Both Sexes					
Age (in years)	Deaths	Population	Crude MR per 1000	US 2000 Std Pop	Expected number of deaths (cross product)
...	...	...	...	...	...
55-59		269,632			
60-64		276,753			
55-64	3,561	388,981	9.15	0.087247	0.80
65-69		186,129			
70-74		131,710			
65-74	5,071	316,674	16.01	0.066037	1.06
75-79		92,967			
80-84		74,275			
75-84	7,718	167,242	46.15	0.044842	2.07
≥85	11,643	86,883	134.01	0.015508	2.08
Crude MR	31,332	3,292,493			
Age-adjusted MR					

WORK IT OUT >>

Age-adjusted Mortality Rate

Age-adjusted mortality rate
 $= \sum \text{Cross products for all age-groups}$
 $= \text{SUM}(F10, F11, F14, F17, F20, F23, F26, F29, F32, F35, F36)$
(spreadsheet program for White Marylanders, both sexes combine)

			White Both Sexes		
Age (in years)	Deaths	Population	Crude MR per 1000	US 2000 Std Pop	Expected number of deaths (cross product)
...	...	...	...	...	...
55-59		269,632			
60-64		276,753			
55-64	3,561	388,981	9.15	0.087247	0.80
65-69		186,129			
70-74		131,710			
65-74	5,071	316,674	16.01	0.066037	1.06
75-79		92,967			
80-84		74,275			
75-84	7,718	167,242	46.15	0.044842	2.07
≥85	11,643	86,883	134.01	0.015508	2.08
Crude MR	31,332	3,292,493			
Age-adjusted MR					7.16

Overall Crude Mortality Rate

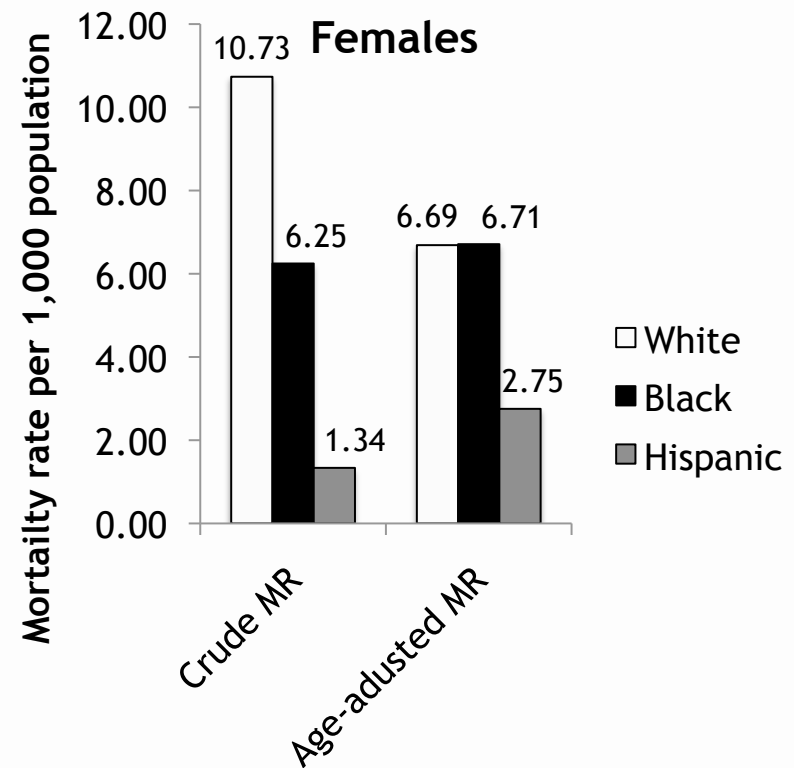
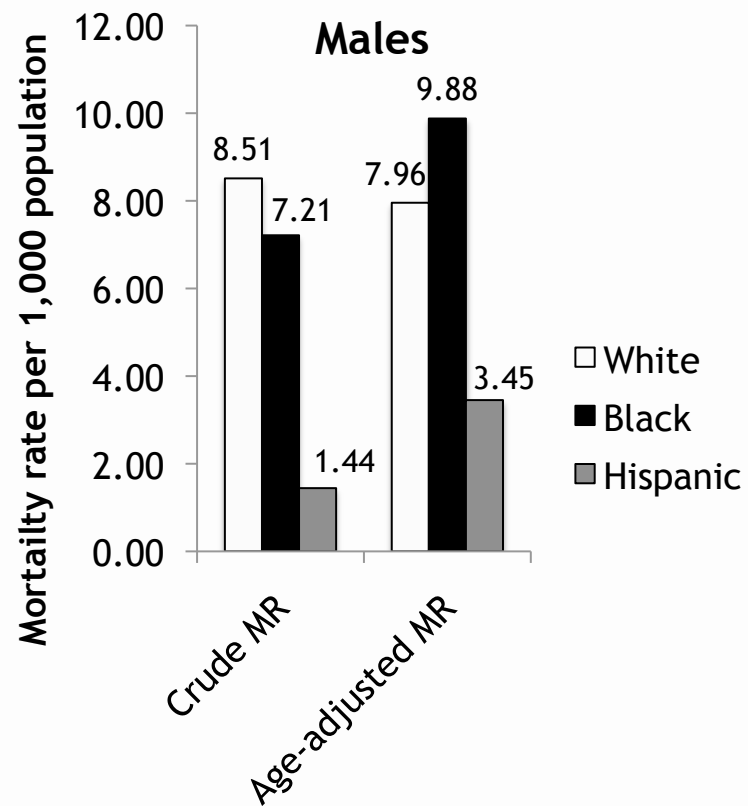
White Both Sexes					
Age (in years)	Deaths	Population	Crude MR per 1000	US 2000 Std Pop	Expected number of deaths (cross product)
...	...	...	...	...	...
55-59		269,632			
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≥85	11,643	86,883	134.01	0.015508	2.08
Crude MR	31,332	3,292,493			
Age-adjusted MR					7.16

WORK IT OUT >>

Overall Crude Mortality Rate

White Both Sexes					
Age (in years)	Deaths	Population	Crude MR per 1000	US 2000 Std Pop	Expected number of deaths (cross product)
...	...	...	...	...	...
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55-64	3,561	388,981	9.15	0.087247	0.80
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75-79		92,967			
80-84		74,275			
75-84	7,718	167,242	46.15	0.044842	2.07
≥85	11,643	86,883	134.01	0.015508	2.08
Crude MR	31,332	3,292,493	9.52		
Age-adjusted MR					7.16

Crude and Age-adjusted Mortality Rates

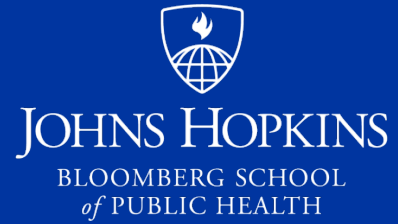


Direct Age Adjustment

- Technical notes
 - ▶ Always report the standard population used
 - ▶ Age-adjusted rates are hypothetical (i.e., not observed data) and useful for comparison purposes
 - ▶ **If there are <25 total deaths within the age groups of the population of interest, indirect age adjustment should be used**
 - ▶ It is not meaningful to age-adjust data for smaller ranges of age groups (i.e., if the total population was 18-24)
 - ▶ If age-specific mortality rates do not have a consistent relationship over time, you should not age adjust (i.e., if MR among younger persons increases over time, but MR among older persons decreases)

Calculations for Direct Age Adjustment

- Crude Mortality Rate =
$$\frac{\text{\# of deaths}}{\text{population count}} * 1,000$$
- Expected number of deaths or cross products =
mortality rate * US 2,000 Std Pop Weight
- Age-adjusted Mortality Rate =
 Σ cross products for all age-groups



Indirect Age Adjustment

Skills-building Example

Indirect Age Adjustment

- Standardized mortality **ratio** (SMR) presents the results of indirect age adjustment
 - ▶ Used less frequently than direct age adjustment
 - ▶ Frequently used in occupational and environmental studies
 - ▶ Useful when age-specific numbers of deaths are small (<25 events in the population of interest) - occurs in small geographical areas
 - ▶ More frequently being used in studies of health inequalities because it is a **measure of RISK** (it directly measures the difference, or gap, between the probability of death in the study population [i.e., population of interest] and standard populations)

- $$\text{SMR} = \frac{\text{\# of observed deaths}}{\text{\# of expected deaths}} * 100$$
 - ▶ **\# of expected deaths** = $\sum(\text{standard age-specific mortality rate}) * (\text{age-specific population counts in the study population})$

Interpretation of the SMR - I

■ Interpretation

- ▶ $SMR > 100$ means the number of observed deaths is **greater than** what would be expected if the study population had the same probability of dying as the standard population (**greater risk**)
- ▶ $SMR = 100$ means the number of observed deaths is the **same as** what would be expected if the study population had the same probability of dying as the standard population (**same risk**)
- ▶ $SMR < 100$ means the number of observed deaths is **less than** what would be expected if the study population had the same probability of dying as the standard population (**less risk**)

Interpretation of the SMR - II

■ Interpretation

► SMR = 278

- The number of observed deaths in the study population is 178% higher than what would be expected if the study population had the same probability of dying as the standard population
- There is a 2.78-fold increase in the number of observed deaths in the study population compared to what would be expected if the study population had the same probability of dying as the standard population
- There is a 178% increase in the **risk** of death in the study population compared to the standard population
- There is a 2.78-fold increase in the **risk** of death in the study population compared to the standard population

Expected Deaths

OVERALL (Hispanic vs. White)

WORK IT OUT >>

Age Group	White Crude MR per 1000	Hispanic Population	Expected Deaths
<1	3.58	11,887	
1-4	0.16	45,679	
5-14	0.07	90,447	
15-24	0.57	81,986	
25-34	0.95	103,347	
35-44	1.77	87,939	
45-54	4.43	58,276	
55-64	9.15	30,953	
65-74	16.01	13,504	
75-84	46.15	6,073	
≥85	134.01	2,283	
Total			0.00

Expected deaths = $\sum (\text{standard age-specific mortality rate}) * (\text{age-specific population counts in the population of interest})$
 = $[\text{White crude MR per 1000} / 1000] * \text{Hispanic Population}$

Expected Deaths

OVERALL (Hispanic vs. White)

Age Group	White Crude MR per 1000	Hispanic Population	Expected Deaths
<1	3.58	11,887	42.58
1-4	0.16	45,679	7.14
5-14	0.07	90,447	6.13
15-24	0.57	81,986	47.02
25-34	0.95	103,347	97.83
35-44	1.77	87,939	155.52
45-54	4.43	58,276	258.13
55-64	9.15	30,953	283.37
65-74	16.01	13,504	216.24
75-84	46.15	6,073	280.26
≥85	134.01	2,283	306.94
Total			1,657.58

Expected deaths = $\sum (\text{standard age-specific mortality rate}) * (\text{age-specific population counts in the population of interest})$
 = $[\text{White crude MR per 1000} / 1000] * \text{Hispanic Population}$

SMR - Hispanics vs. Whites Overall

$$\blacksquare \text{ SMR} = \frac{\text{\# of observed deaths in Hispanics}}{\text{\# of expected deaths in Hispanics}} * 100$$

$$= \frac{741}{1,657.58} * 100$$

$$= 45$$

■ Interpretation

- ▶ The number of observed deaths in Hispanics is 55% lower than what would be expected if Hispanics had the same probability of dying as Whites in Maryland
- ▶ There is a 55% decrease in the risk of death Hispanics compared to Whites in Maryland

SMRs

	Hispanic vs. White	Black vs. White
Overall	45	116
Males	51	138
Females	39	100

Results

- Hispanics have a lower risk of death compared to Whites
- Black men have a higher risk of death than White men
- Black women have the same risk of death as White women

SMRs

■ 95% Confidence Interval for SMR

- ▶ Lower limit (ll) = $SMR - (196 * \sqrt{O/E})$
- ▶ Upper limit (ul) = $SMR + (196 * \sqrt{O/E})$

- ▶ $ll = 45 - (196 * \sqrt{741} / 1,657.58)$
= 42

- ▶ $ul = 45 + (196 * \sqrt{741} / 1,657.58)$
= 48

- ▶ 45 (42, 48)

- ▶ 95% CI does not include 100, the SMR is statistically significantly different than 100

Indirect Age Standardization

■ Technical Notes

- ▶ Indirectly standardized rates compares the mortality rates to the standard population; it is inappropriate to ratio two SMRs

SMR for Hispanic compared to **White** = 45

SMR for Black compared to **White** = 116

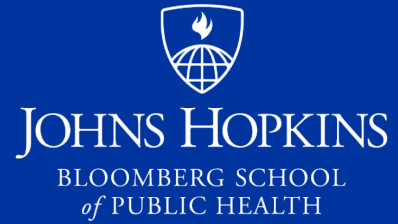
~~Ratio of SMRs = 45 / 116~~

Statistically, it is inappropriate to ratio or statistically compare two SMRs, however, if they have the same standard population, you can compare SMRs to make inferences about priorities, policies, programs, etc

- ▶ It is not meaningful to age-adjust data for smaller ranges of age groups (i.e., if the total population was 18-24)
- ▶ If age-specific mortality rates do not have a consistent relationship over time, you should not age adjust (i.e., if MR among younger persons increases over time, but MR among older persons decreases)

Final Word on Age Adjustment

- There are many ways to adjust (for age or other variables)
 - ▶ Stratified or subgroup analyses
 - ▶ *Direct or indirect standardization of rates*
 - ▶ Life tables
 - ▶ Multivariable statistical analyses



Years of Potential Life Lost (YPLL)

Not all those who wander are lost.
~J.R.R. Tolkien

YPLL

- Years of potential life lost (YPLL) is used to measure premature mortality
 - ▶ YPLL are often presented accompanying MR
 - ▶ I.e., the leading causes of death by age-standardized MR and YPLL

Individual-level data

Σ (End point age - age of decedent who died prior to end point age)

Group-level data

$\Sigma [$ (End point age - midpoint of the age group) *
(number of deaths in the age group)]

- End point age
 - ▶ 75 is used by the NCHS as an approximation of current life expectancy

YPLL

- YPLL can be used to measure **trends**
 - ▶ Can be used to measures **severity** if stratified by age group
- YPLL can be measured at the **local, state, and national** level

YPLL - Individual Level Data

ID	Age at death (A)	End point age (E)	Years Lost (E - A)
1	22	75	53
2	63	75	12
3	89	75	0
4	65	75	10
5	12	75	63
6	34	75	41
7	65	75	10
8	78	75	0
9	45	75	30
10	86	75	0
Total YPLL			191

YPLL - Group Level Data

0 1 2 3 4 5 6 **7** 8 9 10 11 12 13 14 15

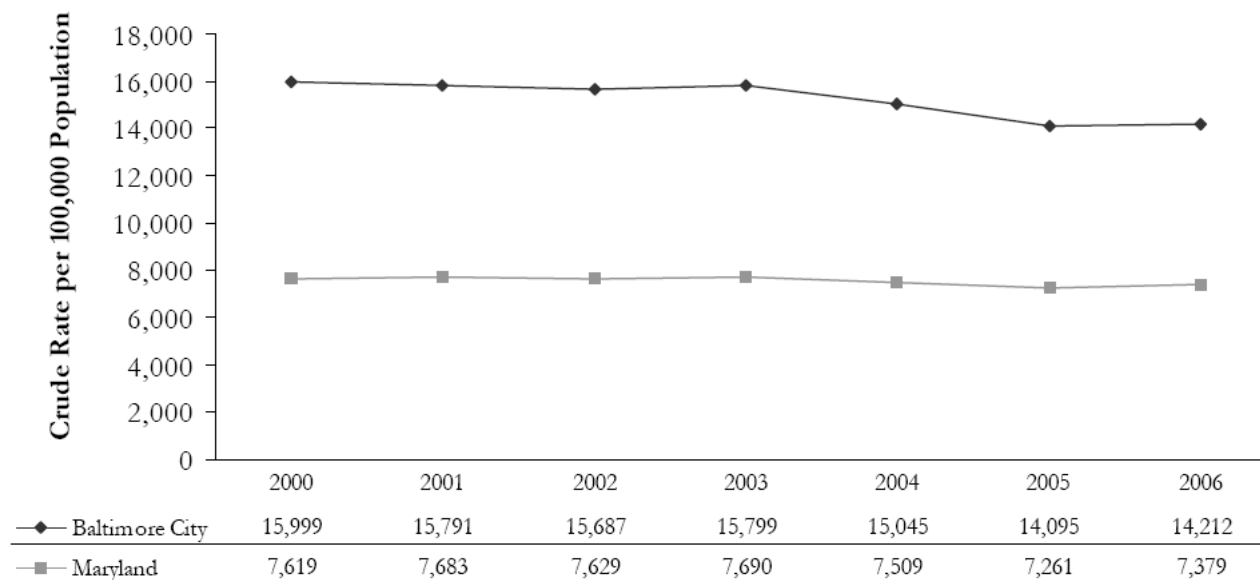
Age group at death	Midpoint of age group at death M	Number of deaths D	End Age A	Years lost (A-M)*D
0 to 14	7.5	6	75	408.0
15 to 24	20	12	75	666.0
25 to 34	30	28	75	1,274.0
35 to 44	40	24	75	852.0
45 to 54	50	29	75	739.5
55 to 64	60	48	75	744.0
65 to 74	70	59	75	324.5
75 to 84	80	79	75	0.0
≥85	85	64	75	0.0
Total YPLL				4,600.0
Total population				65,449
YPLL rate per 1,000 population				70.3

YPLL - Discounting

- YPLL weights a death at a young age more heavily than a death at an older age - sometimes too heavily
 - ▶ An individual who is 24: ages 25-29 are more valuable than ages 60-64
 - ▶ YPLL should be discounted at an annual rate of 3% (US)
- For more information on discounting YPLL, go to <http://www.quantitativeskills.com/sisa/papers/paper6.htm>

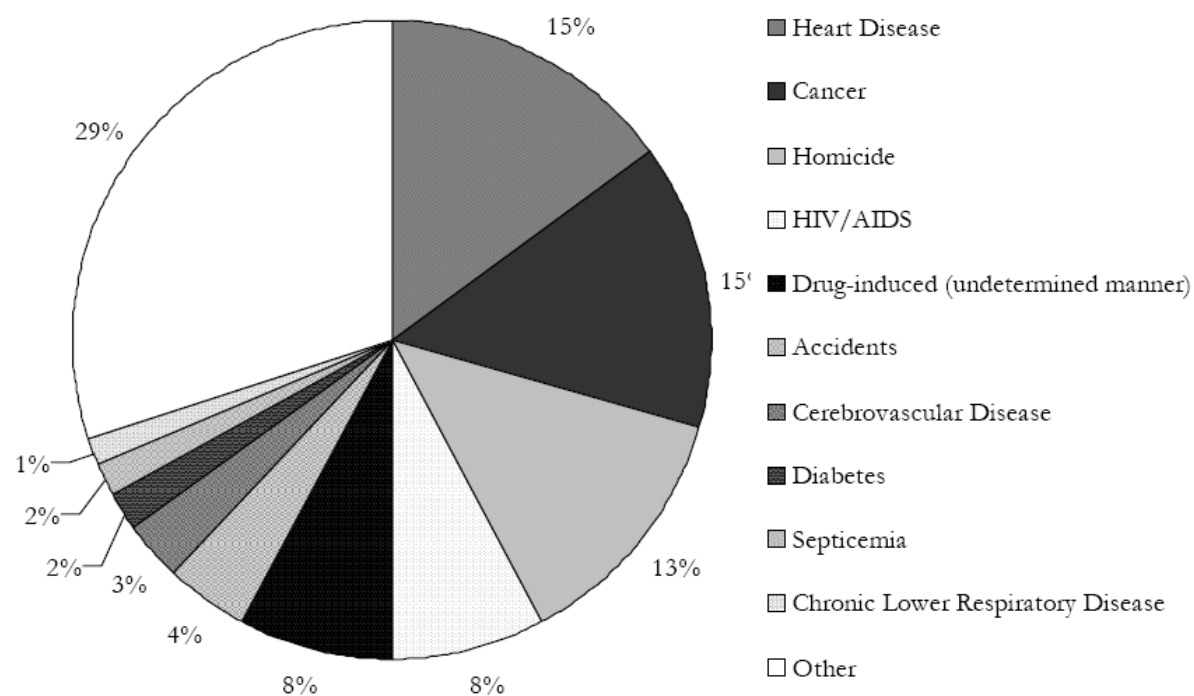
YPLL

Years of Potential Life Lost per 100,000, Baltimore City and Maryland, 2000-2006



Source: Baltimore City Health Department analysis of data from the Maryland Department of Health and Mental Hygiene, Vital Statistics Administration: Baltimore City Vital Statistics Profile and the Maryland Vital Statistics Reports. Years of Potential Life Lost calculated using cut-off of age 75 years.

Years of Potential Life Lost for the Top 10 Causes of Death, Baltimore City, 2006



Source: Baltimore City Health Department analysis of data from the Maryland Department of Health and Mental Hygiene, Vital Statistics Administration: 2006 Baltimore City Profile. Years of Potential Life Lost calculated using cut-off of age 75 years. Drug-Induced Deaths of undetermined manner - this category does not include deaths due to accident, suicide or homicide ICD-10 codes Y10-Y14. See technical notes for more information on cause of death categories.

Life Expectancy

*Infant mortality and life expectancy are
reasonable indicators of general well-being
in a society.*
~P.J. O'Rourke

Life Expectancy

- Life expectancy (e_x) is the average years of life remaining at a given age (x) if the currently prevailing age-specific mortality rates continue throughout the life span
 - ▶ e_0 = the average years of life remaining at birth if the newborn experiences the currently prevailing age-specific mortality through his/her lifespan
 - ▶ Life expectancy can be used to monitor **trends**
 - ▶ Life expectancy can be measured at the **local, state, and national** level

Life Expectancy

- Life expectancy is depicted using life tables
 - ▶ Often stratified by sex

- Types of life tables
 - ▶ Cohort
 - represents mortality of a particular cohort followed through time
 - ▶ Period or Current (PEM1)
 - what would happen to a hypothetical cohort if it experienced throughout its entire life the mortality conditions of a particular time period

Life Table - National Vital Statistics Reports

Table 1. Life table for the total population: United States, 2011

Age	Probability of dying between ages x to x + 1	Number surviving to age x	Number dying between ages x to x + 1	Person- years lived between ages x to x + 1	Total number of person-years lived above age x	Expectation of life at age x
	q_x	l_x	d_x	L_x	T_x	e_x
	q_x	l_x	d_x	L_x	T_x	e_x
0-1	0.006059	100,000	606	99,470	7,870,915	78.7
1-2	0.000415	99,394	41	99,374	7,771,445	78.2
2-3	0.000264	99,353	26	99,340	7,672,071	77.2
3-4	0.000208	99,327	21	99,316	7,572,731	76.2
4-5	0.000167	99,306	17	99,298	7,473,415	75.3
5-6	0.000151	99,289	15	99,282	7,374,117	74.3

Bold numbers are inserted and not from the formulas noted

Life Tables - National Vital Statistics Reports

- Age
 - ▶ “20-21” means the 1-year interval between the 20th and 21st birthdays.
- q_x = Probability of dying between x and $x + 1$
 - ▶ q_x = # of deaths / population count
 - ▶ This column forms the basis of the life table; all subsequent columns are derived from it
- l_x = Number surviving to age x
 - ▶ $l_x = l_{x-1} * (1 - q_{x-1})$
 - ▶ The number of persons from the original synthetic cohort of 100,000 live births, who survive to the beginning of each age interval
- d_x = Number dying between age x and $x+1$
 - ▶ $d_x = l_x * q_x$
 - ▶ Number dying in each successive age interval out of the original 100,000 live births

Arias E, United States Life Tables, 2011. National Vital Statistics Reports 2015;64(11):1-62.

Life Tables - National Vital Statistics Reports

- L_x = Person years lived between ages x and $x+1$
 - ▶ $L_x = l_x - \frac{1}{2}d_x$
 - ▶ 98,881 in the age interval 20-21 is the total number of years lived between the 20th and 21st birthdays by the 98,632 males who reached their 20th birthday out of 100,000 males born alive

- T_x = Total number of person years lived above age x
 - ▶ $T_x = \sum L_x$
 - ▶ Total number of person-years that would be lived after the beginning of the age interval x to $x + 1$ by the synthetic life table cohort

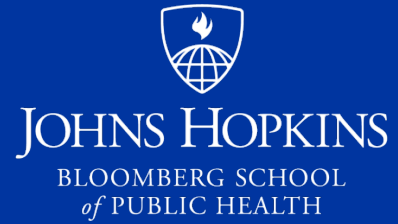
- e_x = Life expectancy or expectation of life at age x
 - ▶ $e_x = T_x / l_x$
 - ▶ Average number of years remaining to be lived by those surviving to that age on the basis of a given set of age-specific rates of dying

Arias E, United States Life Tables, 2011. National Vital Statistics Reports 2015;64(11):1-62.

Life Expectancy

Table 3.1
Countries with the highest and lowest life expectancy at birth (in years), by sex, 2015

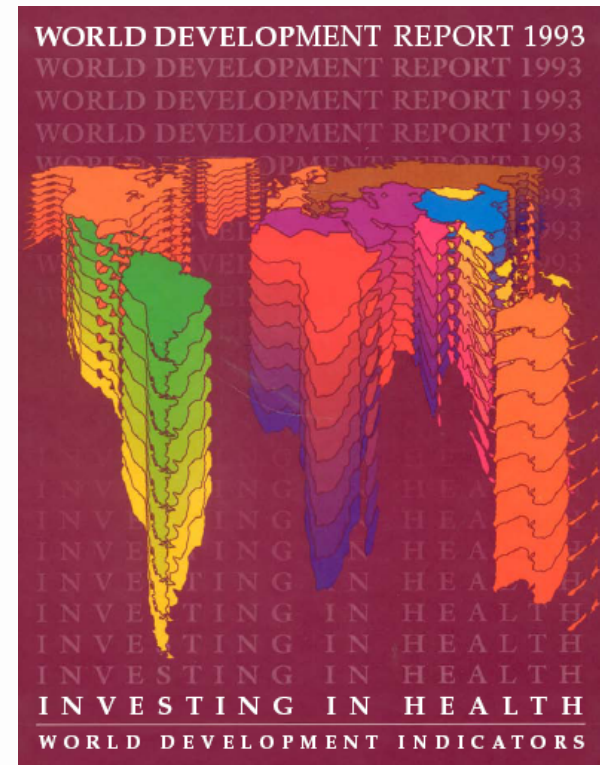
Male		Female	
Country	Years	Country	Years
Highest		Highest	
Switzerland	81.3	Japan	86.8
Iceland	81.2	Singapore	86.1
Australia	80.9	Spain	85.5
Sweden	80.7	Republic of Korea	85.5
Israel	80.6	France	85.4
Japan	80.5	Switzerland	85.3
Italy	80.5	Australia	84.8
Canada	80.2	Italy	84.8
Spain	80.1	Israel	84.3
Singapore	80.0	Iceland	84.1
Lowest		Lowest	
Lesotho	51.7	Chad	54.5
Chad	51.7	Côte d'Ivoire	54.4
Central African Republic	50.9	Central African Republic	54.1
Angola	50.9	Angola	54.0
Sierra Leone	49.3	Sierra Leone	50.8



Disability-adjusted Life Years (DALY)

History of DALYs

- Developed as an input into the World Bank's "*World Development Report 1993: Investing in Health*"
 - ▶ DALYs are used in prioritization and decision making
 - ▶ Most widely used quality-adjusted measure of disease burden



DALYs

- Disability-adjusted life years (DALY)
 - ▶ DALYs “extends the concept of potential years of life lost due to premature death...to include equivalent years of ‘healthy’ life lost by virtue of being in states of poor health or disability”
 - Combines data on mortality and non-fatal health outcomes
 - ▶ DALYs measure the difference between the current situation and an ideal situation where everyone lives up to the age of the standard life expectancy in perfect health
 - Stratified by gender due to difference in life expectancy

"Death and DALY estimates for 2004 by cause for WHO Member States: Persons, all ages" (xls). World Health Organization. 2002.
http://www.who.int/entity/healthinfo/global_burden_disease/gbddeathdalycountryestimates2004.xls.

DALYs

- ▶ Interpreted as the number of lost healthy years
 - The public health goal is to reduce DALYs to a minimum
- ▶ DALYs can be used to measure **trends and severity** of illness
- ▶ DALYs can be measured at the **national** level

"Death and DALY estimates for 2004 by cause for WHO Member States: Persons, all ages" (xls). World Health Organization. 2002.
http://www.who.int/entity/healthinfo/global_burden_disease/gbddeathdalycountryestimates2004.xls.

DALYs Calculation

$$\text{DALY} = \text{YLL} + \text{YLD}$$

Years of life lost due to premature mortality (YLL)
 $= N * L$

N is the # of deaths

L is the standard life expectancy at age of death

Years lived with disability (YLD) $= I * DW * L$

I is the # of incident cases

DW is the disability weight (0=perfect health, 1=dead)

L is the average duration of disability

DALY - calculation By Cause, Age & Sex

$$\text{DALY} = \text{YLL} + \text{YLD}$$

$$\text{YLL} (c,a,s) = N(c,a,s) * L(a,s)$$

$N(c,a,s)$ is the number of deaths due to cause c
for age a and sex s

$L(a,s)$ is the standard loss function in years for
age a and sex s

$$\text{YLD} (c,a,s) = I(c,a,s) * DW(c,a,s) * L(c,a,s)$$

$I(c,a,s)$ is the number of incident cases for cause
 c , age a , sex s

$DW(c,a,s)$ is the disability weight (0=perfect
health, 1=dead) for cause c , age a , sex s

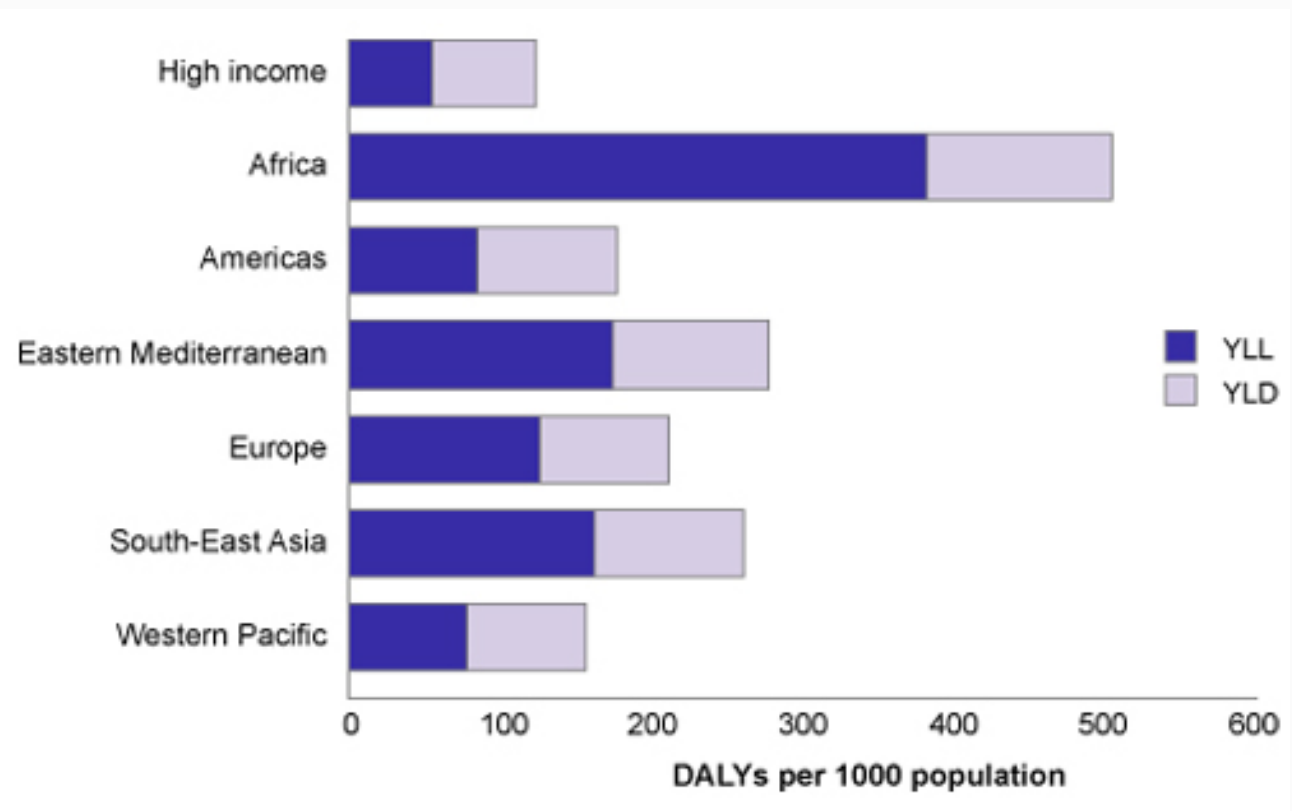
$L(c,a,s)$ is the average duration in years of the
case until remission or death

Disability Weights

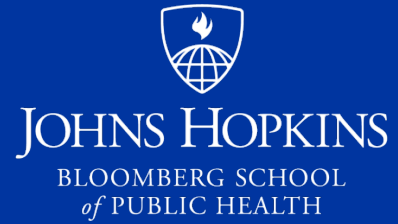
Disease or sequelae	Mean disability weight (untreated form)	Mean disability weight (treated form)
AIDS	0.50	0.50
Infertility	0.18	0.18
Diarrhoea disease, episodes	0.11	0.11
Measles episode	0.15	0.15
Tuberculosis	0.27	0.27
Malaria, episodes	0.20	0.20
Trachoma, blindness	0.60	0.49
Trachoma, low vision	0.24	0.24
Lower respiratory tract infection, episodes	0.28	0.28
Lower respiratory tract infection, chronic sequelae	0.01	0.01
Cancers, terminal stage	0.81	0.81
Diabetes mellitus cases (uncomplicated)	0.01	0.03
Unipolar major depression, episodes	0.60	0.30
Alcohol dependence syndrome	0.18	0.18
Parkinson disease cases	0.39	0.32
Alzheimer disease cases	0.64	0.64
Post-traumatic stress disorder	0.11	0.11
Angina pectoris	0.23	0.10
Congestive heart failure	0.32	0.17
Chronic obstructive lung disease, symptomatic cases	0.43	0.39
Asthma, cases	0.10	0.06
Deafness	0.22	0.17
Benign prostatic hypertrophy	0.04	0.04
Osteoarthritis, symptomatic hip or knee	0.16	0.11
Brain injury, long-term sequelae	0.41	0.35
Spinal cord injury	0.73	0.73
Sprains	0.06	0.06
Burns (>60%) – long term	0.25	0.25

^a Adapted from Murray & Lopez (1996).

DALY



World Health Organization (
http://gamapserver.who.int/gho/static_graphs/mbd/dalys_regional.jpg)



Quality-Adjusted Life Years (QALY)

History of QALYs

- QALYs were developed in the 1960s to facilitate health economic analyses
- Klarman HE, Francis JO, Rosenthal GD. 1968. Cost-effectiveness analysis applied to the treatment of chronic renal disease. *Med Care* 1968;6:48-54.

QALY

- Quality-adjusted Life Year (QALY)
 - ▶ Combines data on the quality and quantity of life lived
 - ▶ It is used as a measure of benefit in economic evaluations
 - QALYs quantifying the cost of a healthy year of life lived
 - An intervention is cost effective if it costs <\$50,000/QALY
 - ▶ QALY: 1=perfect health, 0=death
 - ▶ QALYs can be used to measure **trends** and **severity** of illness
 - ▶ QALYs can be measured at the **national** level

- Many ways to calculate QALYs
 - ▶ Incorporates
 - Expert opinion on quality of life
 - Perspective of those who are living a life with health impairment (data collected via survey)

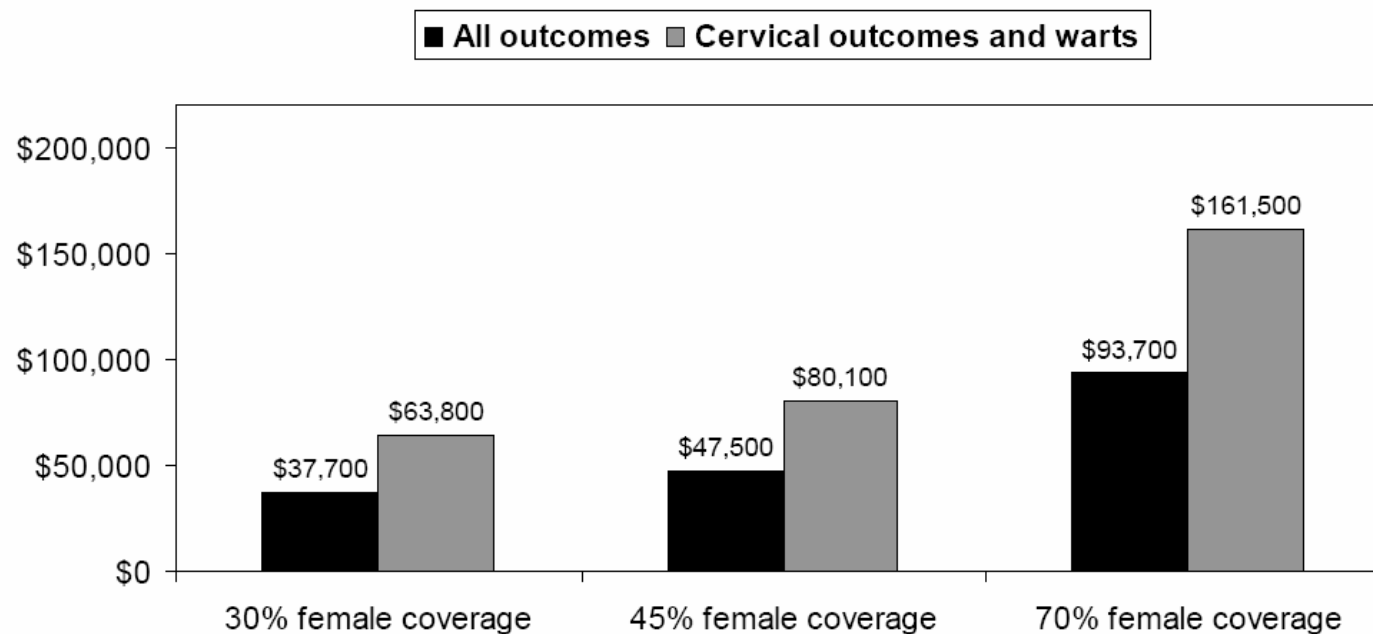
Interpretation of QALY

- Example: QALY = 0.75
 - ▶ Remember: QALY: 1=perfect health, 0=death
 - ▶ 25% reduction in the quality of life

- QALYs calculated at a **population** level
 - ▶ Usually the QALYs are summed over the population and then costs are incorporated in a number of ways
 - ▶ Interpreted as the number of healthy years lived

QALY

Incremental cost per QALY of male HPV vaccination



Chesson, H. Advisory Committee on Immunization Practices meeting, October 21, 2009. (<http://www.cdc.gov/vaccines/recs/acip/downloads/mtg-slides-oct09/02-8-hpv.pdf>)



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Summary

Summary

- Mortality rates are age adjusted for comparison purposes
 - ▶ Mortality rates can be age adjusted using direct or indirect adjustment methods
 - ▶ Crude rates should also be presented with adjusted rates
- YPLL describes the years of potential life lost, and is heavily influenced by deaths at young ages
- Period life tables are used to calculate life expectancy
 - ▶ Life tables are usually stratified by sex

Summary

- DALYs combine premature death and loss of healthy life
 - ▶ DALYs are usually calculated differently for males and females due to the difference in life expectancy
 - ▶ Interpreted as the number of *lost healthy years*
 - ▶ Disability weights: 0=perfect health, 1=death

- QALYs measure years of healthy life discounted for lower quality of life
 - ▶ Interpreted as the number of *healthy years lived*
 - ▶ QALY: 1=perfect health, 0=death